

Female Reservation and the Qualifications of Local Elected Officials in India

Gaurav Sood

Working draft

Abstract

Reserving a local office for women changes who can win it. We compare winners of women-reserved and open seats in rural councils in Bihar, Uttar Pradesh, Rajasthan and Kerala and in the municipal councils of Mumbai and Delhi. Among village heads in Bihar and Uttar Pradesh, reserved-seat winners are 7 to 15 percentage points less likely to hold a university degree. Among Kerala's village council members and in the two cities, graduate shares are about the same. Economic position differs in every village, block and city office with occupation records: reserved-seat winners are 22 to 55 points more likely to report no occupation, and in Rajasthan they declare about 47% fewer assets. In Mumbai and Delhi they are 11 to 27 points less likely to face a pending criminal case, a record that covers the officeholder but not the male relatives who often act for her. Published comparisons of village heads agree with the rural results: across five independent samples, reserved-seat heads have 0.88 standard deviations less schooling.

1 Introduction

Reserving a seat for women changes who can win it. This paper asks whether the winners of reserved seats are less qualified than the winners of open seats. Open seats can and do elect women, so the comparison is between seat types, not between women and men.

The answer depends on where one looks. Among village heads in Bihar and Uttar Pradesh, reserved-seat winners are much less likely to hold a degree. Among Kerala's village council members and the councillors of Mumbai and Delhi, they are not. Setting accounts for most of the variation in the education gap across the six settings we study. Economic position is different: in every village, block and city office with occupation records, including Kerala's and Delhi's, reserved-seat winners far more often report no occupation of their own. In the two cities they are also much less likely to face pending criminal cases.

The rural results match the published evidence. Six studies compare the schooling of village heads in reserved and open seats, and all find less schooling under reservation; pooled, the gap is close to one standard deviation. The pattern in occupation fits a separate body of work showing that male relatives often stand in for women elected under quotas.

2 Why these attributes matter

Each attribute candidates declare when they file to run has a claim to shaping how officials govern. Village heads prepare budgets, run public works and allocate benefits, work that requires reading

documents, applying rules and dealing with officials (Auerbach et al. 2025, 708–10; Joshi et al. 2025, 56). In South India, more educated council heads target benefits better to landless households and secure fewer for their own (Besley et al. 2007, Tables 4–5), and graduates who narrowly win Indian state assembly seats raise night-time-light growth in their constituencies (Jain et al. 2022, Table 2), though college-educated leaders do no better across national leaders, US legislators and Brazilian mayors (Carnes and Lupu 2016). Narrowly electing a state legislator who faces criminal charges lowers growth and road provision (Prakash et al. 2019). Age shapes incentives: younger Italian mayors facing reelection run larger electoral budget cycles (Alesina et al. 2019, Tables 5–7).

Occupation and assets measure economic independence. A winner with no occupation and few assets of her own depends on her household, which is one route by which husbands and fathers-in-law come to exercise the office (Karekurve-Ramachandra and Sood 2026, 1–2; Heinze et al. 2025).

3 Existing evidence

Six studies compare the schooling of women-reserved and open-seat village heads (Table E.1). All find less schooling under reservation: about two to three years where schooling is measured in years (Chattopadhyay and Duflo 2001, Table 3; Beaman et al. 2009, Table II; Ban and Rao 2007, Table 9), lower completion of secondary school in a national village panel and in Andhra Pradesh (Deininger et al. 2011, Table 3; Afridi et al. 2013, Table 2), and 0.92 standard deviations at Bihar’s reservation threshold (Bamezai et al. 2026, Table 7). On a common scale, the standardized difference in means, and counting the two studies of the same Birbhum councils once, the five samples pool to -0.88 (95% CI -1.01 to -0.76). They differ from one another no more than sampling error predicts.

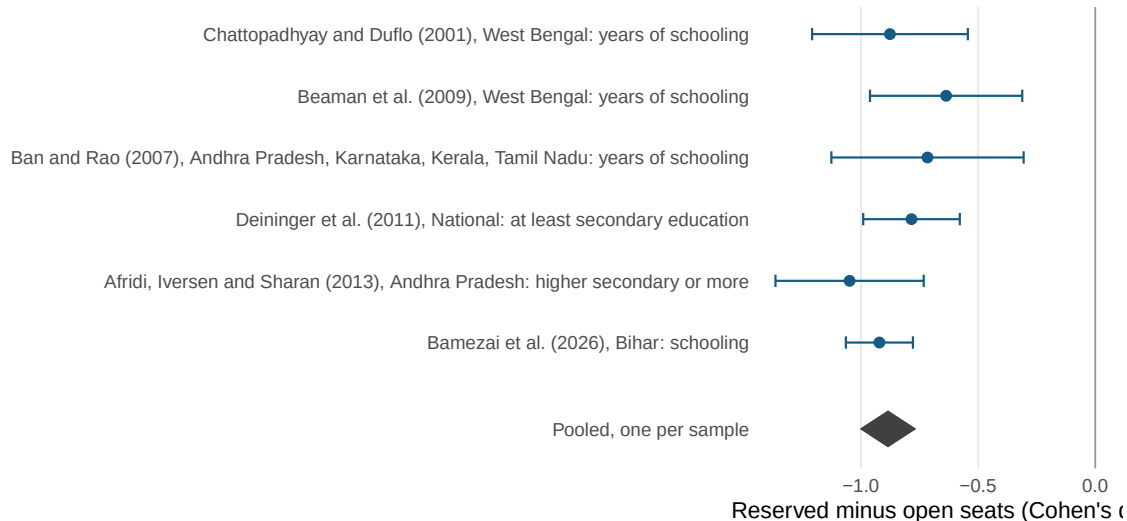


Figure 1: Published comparisons of schooling among elected village heads, women-reserved minus open seats, as standardized differences with 95% confidence intervals. Years of schooling are divided by the pooled within-group standard deviation; shares of heads with secondary schooling are converted through the log odds ratio; Bamezai et al. report standard-deviation units. The pooled estimate is a random-effects mean over five independent samples, with the Birbhum councils entered once through Chattopadhyay and Duflo.

Measured against other women in their villages, reserved-seat leaders are positively selected: in Bihar they are more educated than the average woman (Bamezai et al. 2026, sec. 7.2.2), in South

India they come from wealthier and better-informed households than eligible women (Ban and Rao 2007, Table 9), and in Karnataka selection among women is stronger than among men (Mori et al. 2025). In Bihar, the husbands of reserved-seat winners are also less educated than open-seat winners, by 0.36 standard deviations (Bamezai et al. 2026, Table 7). Two further studies compare different groups: reserved women against men in all other seats in Karnataka (Raabe et al. 2009, Table 4), and candidates’ standing in their village’s education distribution (Mori et al. 2022, Table 3).

4 Data and empirical strategy

4.1 Outcomes and samples

The main education outcome is whether the winner holds a university degree; the appendix also reports illiteracy. Unknown qualifications stay missing. Occupation is recorded in Rajasthan, Kerala and Delhi. A winner counts as having no occupation when the entry is nil, unemployed, homemaker or student; pensioners and every named trade or job count as an occupation, and the appendix lists every label with its code. Declared assets come from the nomination affidavits of Rajasthan’s 2020 and Uttar Pradesh’s 2021 village heads. Pending criminal cases come from MyNeta’s affidavit summaries in Mumbai and Delhi and from Uttar Pradesh’s 2021 affidavits.

4.2 Specification and inference

For each state, election and office, we estimate

$$Y_i = \alpha + \beta R_i + \gamma_{g(i)} + \delta_{c(i)} + \epsilon_i,$$

where R_i indicates a seat reserved for women, $\gamma_{g(i)}$ is a block effect (a district effect where blocks are not recorded throughout) and $\delta_{c(i)}$ is an effect for the seat’s caste-reservation class. For binary outcomes, 100β is the reserved-minus-open difference in percentage points among seats in the same block and caste class. It is the effect of reservation if reservation is as good as randomly assigned within those cells. Standard errors are clustered by block or district, and 95% intervals use a t distribution with one fewer degree of freedom than there are clusters; Kerala’s ward and district-member models have only 14 clusters. Mumbai pools its 2012 and 2017 councils with administrative-ward and council effects, clustered by electoral ward, and has no caste-reservation control. Delhi uses assembly-constituency and caste-reservation effects in each election, clustered by assembly constituency.

The administrative records come from state election repositories through local_elections, with input hashes, winner rules, exclusions and coding dictionaries saved in the repository. Bihar’s winner is the highest vote-getter, an uncontested candidate, or the winner of a tie drawn by lot; seats whose results form cannot identify a winner are excluded. Kerala’s records are of elected ward members, and their reservation refers to the ward even for members who later lead the council.

5 Rural panchayats

5.1 Village heads

Rural village heads: graduate or above

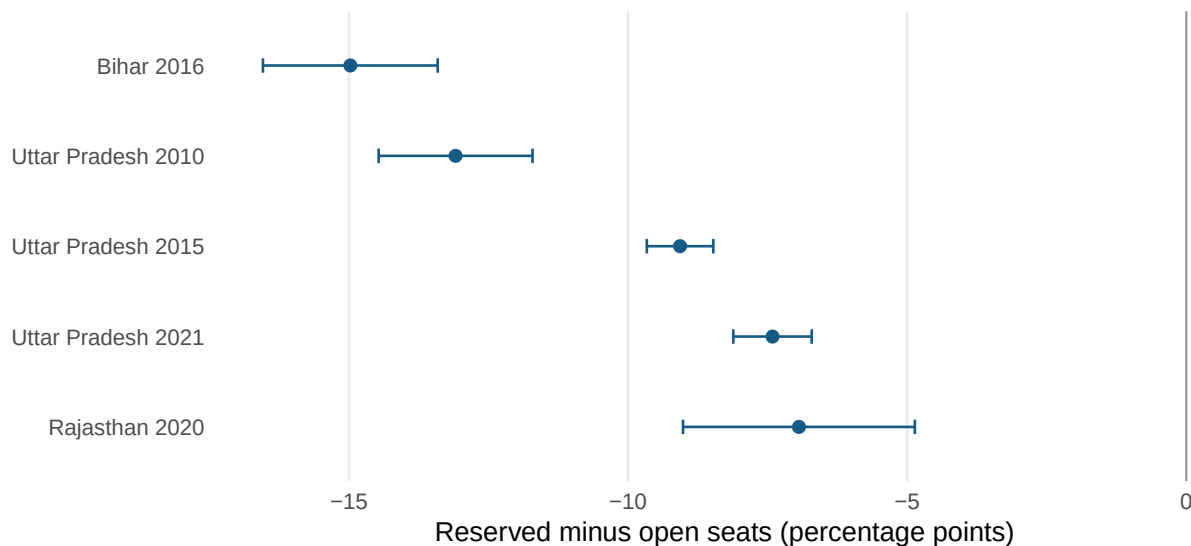


Figure 2: Graduate-share differences among rural village heads, women-reserved minus open seats, with geography and caste-reservation effects. Bars are 95% confidence intervals clustered by geography. UP 2010 uses districts; other rows use blocks. Unclassified education, ambiguous winners and unknown reservation are excluded.

In Uttar Pradesh, the graduate-share gap is -13.1 percentage points (95% CI -14.5 to -11.7) in 2010, -9.1 percentage points (95% CI -9.7 to -8.5) in 2015 and -7.4 percentage points (95% CI -8.1 to -6.7) in 2021. In Rajasthan it is -6.9 percentage points (95% CI -9.0 to -4.9) among heads whose qualification can be classified. Many Rajasthan qualifications are recorded only as “Other”; if each could be either a degree or not, the all-winner difference lies anywhere between -29.7 and 31.5 points.

5.2 Bihar’s other offices

Bihar’s 2016 records cover 210,717 elected seats across six village, block and district offices. The graduate-share gap is negative in every office.

Bihar 2016: graduate or above

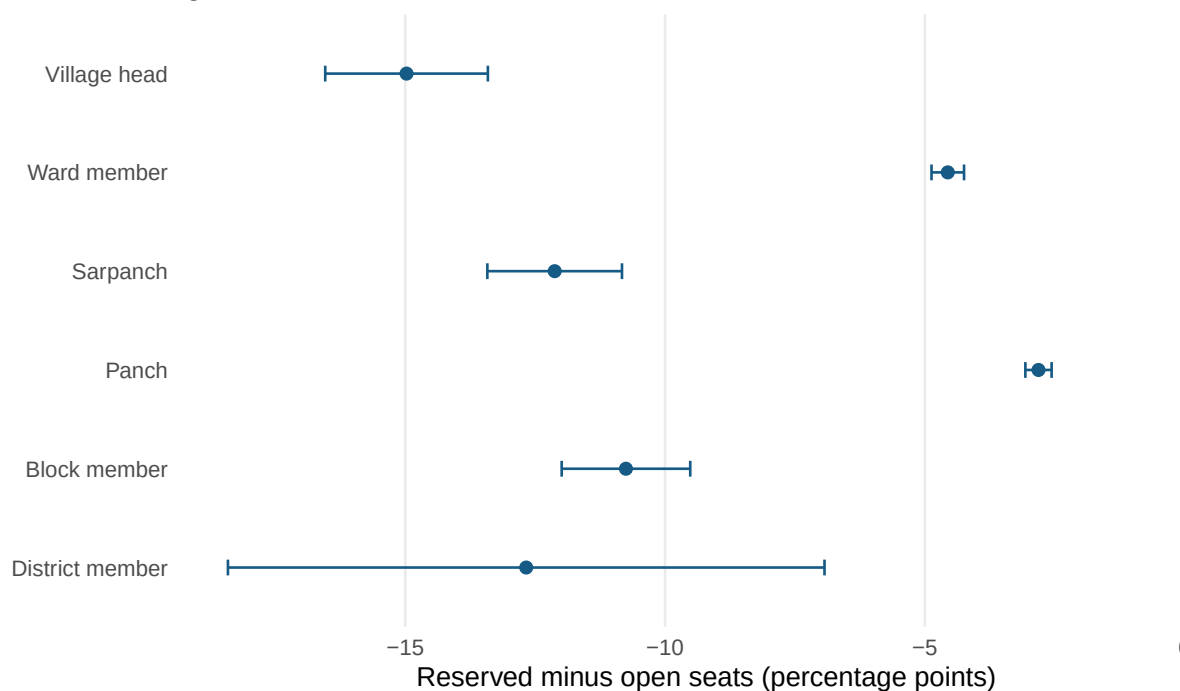


Figure 3: Bihar 2016: graduate-share differences by office, women-reserved minus open seats, with block and caste-reservation effects (district effects for district council members). Bars are 95% confidence intervals clustered at that level.

Bihar's 2021 records add 244,474 elected seats with age but not education. Reserved-seat village heads are 3.2 years younger in 2021 (95% CI 2.7 to 3.6), against 3.6 years in 2016. Reservation for 2021 comes from the State Election Commission's feed for the 2021–2026 term, which matches the Commission's count of women-reserved village-head seats.

5.3 Kerala

Kerala: graduate or above

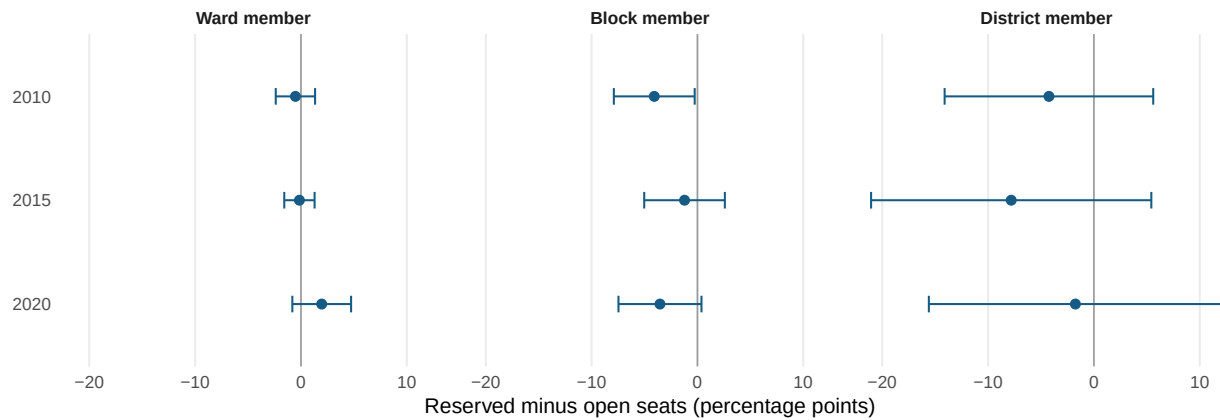


Figure 4: Rural Kerala: graduate-share differences by election and office, women-reserved minus open seats, with caste-reservation effects. Block members use block effects and clustering; village and district members use districts. Bars are 95% confidence intervals.

Among Kerala’s village ward members, the graduate-share differences are -0.5 percentage points (95% CI -2.4 to 1.3), -0.1 percentage points (95% CI -1.6 to 1.3) and 2.0 percentage points (95% CI -0.8 to 4.7). The same office in Bihar shows a gap of 4.6 points, so the difference from the northern states is not only a difference between heads and members. Kerala records education as free text; the coding recognizes degrees and school qualifications and leaves incomplete or unrecognized entries missing.

6 Urban councils

6.1 Mumbai

Mumbai’s councillor records come from the Praja Foundation’s report cards and affidavits, as deposited by Karekurve-Ramachandra and Lee (2023). Each councillor enters once per term.

Urban: Mumbai councillors, 2012 and 2017

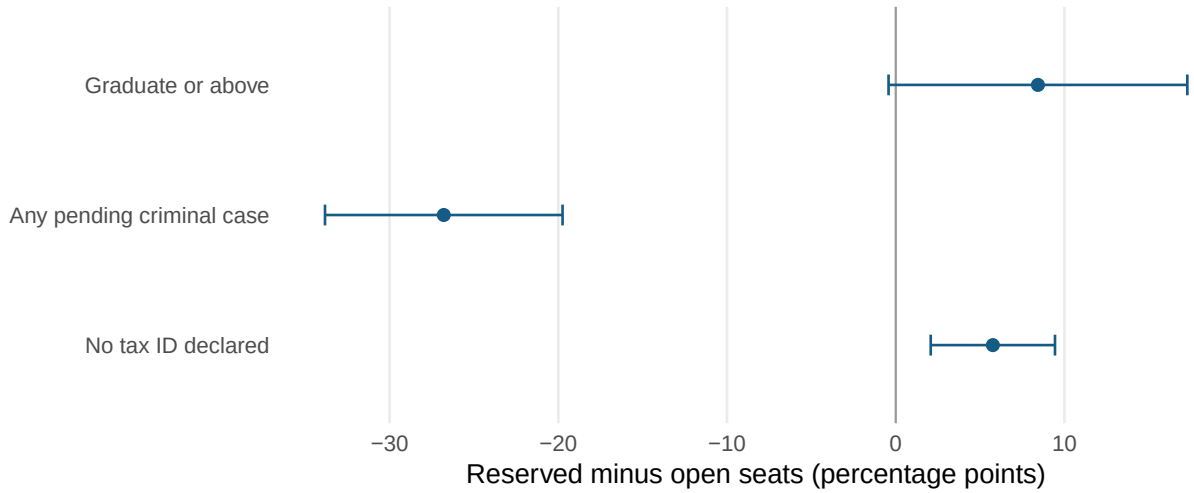


Figure 5: Mumbai councillors in the 2012 and 2017 councils: women-reserved minus open seats in graduate share, any pending case and declaring no tax ID (PAN), with administrative-ward and council effects. Bars are 95% confidence intervals clustered by electoral ward.

The graduate-share difference is 8.4 percentage points (95% CI -0.4 to 17.3) and the pending-case difference is -26.8 percentage points (95% CI -33.8 to -19.7).

6.2 Delhi

Delhi adds the 2012, 2017 and 2022 elections. Winners and assembly mapping for 2012 and 2017 come from Goyal's replication deposit (Goyal 2023); for 2022, winners come from the OpenCity compilation of Lok Dhaba results and reservation from the State Election Commission's gazette (Vaidyanathan 2023; State Election Commission, NCT of Delhi 2022, Annexure B). The 2012 reservation labels match the official schedule (State Election Commission, NCT of Delhi 2012, Annexures I–III). Qualifications and pending cases come from each winner's MyNeta profile (Association for Democratic Reforms 2026); convictions are excluded from pending cases.

Urban: Delhi councillors

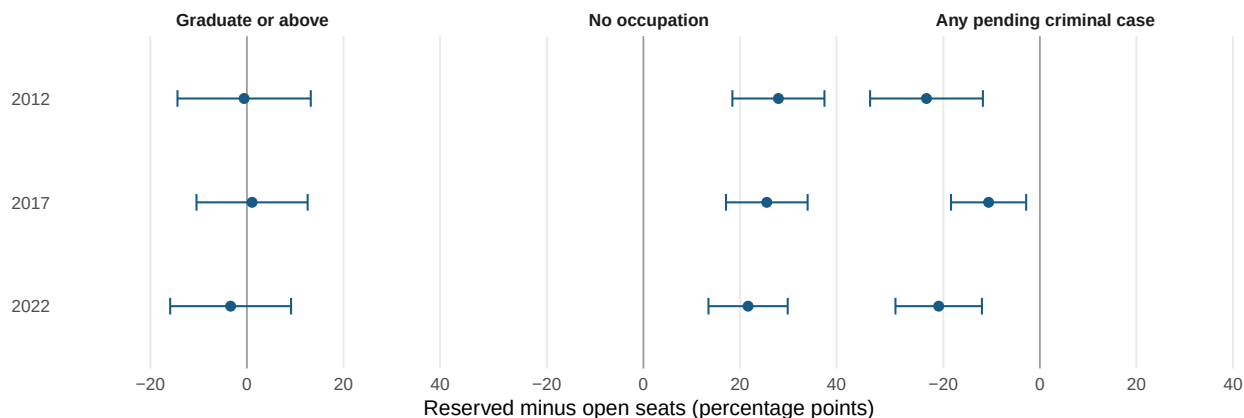


Figure 6: Delhi councillors by election: women-reserved minus open seats in graduate share, reporting no occupation and any pending case, with assembly-constituency and caste-reservation effects. Bars are 95% confidence intervals clustered by assembly constituency.

The graduate-share differences are -0.6 percentage points (95% CI -14.4 to 13.2), 1.1 percentage points (95% CI -10.4 to 12.6) and -3.4 percentage points (95% CI -15.9 to 9.1) in 2012, 2017 and 2022. The pending-case differences are -23.5 percentage points (95% CI -35.2 to -11.8), -10.6 percentage points (95% CI -18.4 to -2.8) and -21.0 percentage points (95% CI -29.9 to -12.0), and stay negative however the missing case records are filled in.

We do not use the education recorded in the older Delhi files, which is also the coding in Karekurve-Ramachandra and Lee’s deposit (Sood 2022; Karekurve-Ramachandra and Lee 2020b). For the 2012 winners, its graduate status agrees with the winners’ own MyNeta profiles 54% of the time, about what shuffling the values across winners gives (53%), while the same records’ declared assets match 97% of the time. The 2017 education labels agree with the profiles for 261 of 262 winners. The Mumbai deposit agrees with MyNeta on both education and pending cases.

7 Occupation, assets and criminal records

Reserved-seat winners are far more likely to report no occupation of their own, including where their education matches that of open-seat winners. In Rajasthan, 51 points more reserved-seat heads report no occupation (95% CI 48 to 54). Among Kerala’s ward members the gap is 41 to 55 points across the three elections, and among Delhi’s councillors 22 to 28 points. Among Kerala’s district council members the gap is smaller, 0 to 15 points. Mumbai records no occupation, but reserved-seat councillors there are 5.8 percentage points (95% CI 2.1 to 9.4) more likely to declare no tax ID, all of it in the 2012 council. Counting “social worker” as missing rather than as an occupation moves no estimate by more than 6 points.

Reserved-seat heads also declare fewer assets: about 47% less in Rajasthan and 9% less in Uttar Pradesh. Criminal records are rare among Uttar Pradesh’s 2021 village heads, 0.18% of those with a known record, and do not differ by seat type (-0.04 points, 95% CI -0.12 to 0.04). The appendix reports every estimate.

8 Discussion

The education gap follows the setting. Pooling this paper’s estimates with elections nested in settings, reserved-seat village heads in the northern states are 10.5 points less likely to hold a degree (95% CI 5.2 to 15.8), against a difference of 1.4 points (-4.9 to 7.6) among Kerala’s ward members and the two cities’ councillors. Distinguishing the two groups cuts the unexplained spread across settings from 7.3 to 2.6 points. Office, data source and election years vary along with setting, and the second group is one state and two cities. Replacing Bihar’s village heads with its ward members, the office we observe in Kerala, still leaves a northern gap of 7.5 points.

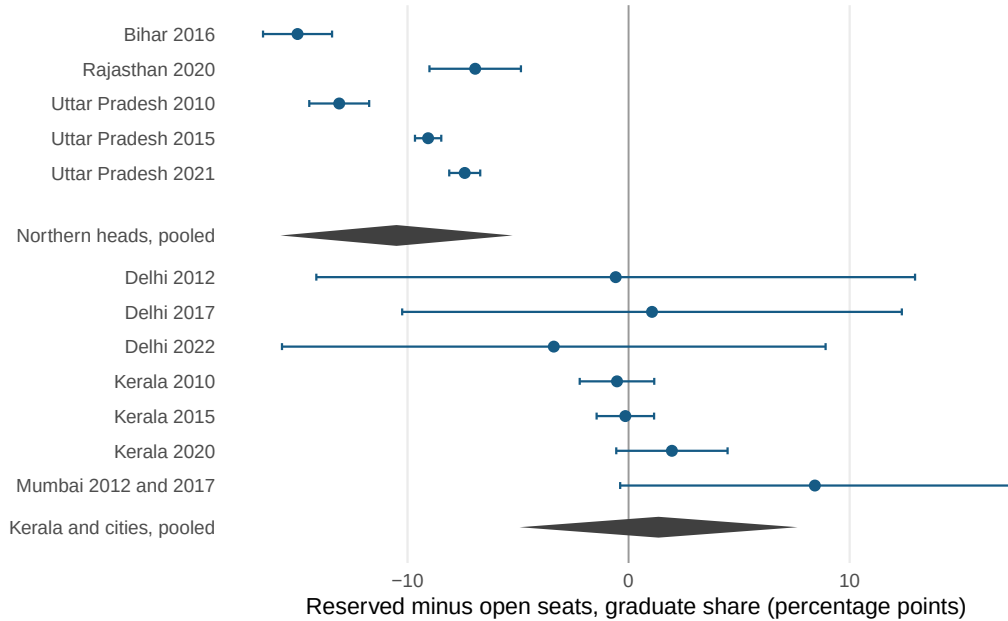


Figure 7: This paper’s graduate-share differences, women-reserved minus open seats, with 95% confidence intervals. Pooled estimates come from a random-effects model with elections nested in settings and an indicator for the northern village-head settings.

Where reserved-seat winners match open-seat winners on education, they still differ in economic independence. That is consistent with evidence that male relatives often exercise the office. In a Rajasthan phone audit, 9% of answered calls to reserved-seat village heads reached the elected woman, against 76% in open seats and 31.7% in urban Jaipur (Karekurve-Ramachandra and Sood 2026, 1–2). Women elected through quotas participate less in council decisions (Heinze et al. 2025), and surveys in two states find male family members prominent in everyday governance (Amar et al. 2025). For the same reason, the lower rate of pending cases among urban reserved-seat councillors may overstate the change in the office’s criminal exposure: affidavits list the officeholder’s own cases, not her relatives’, and Karekurve-Ramachandra and Lee (2020a, 11) note that many Mumbai quota women have male relatives who face charges.

Less-qualified winners need not govern worse. In Rajasthan and Uttar Pradesh, lottery-assigned reservation has no detectable effect on the scale or content of public works (Green et al. 2026).

References

- Afridi, Farzana, Vegard Iversen, and M. R. Sharan. 2013. *Women Political Leaders, Corruption and Learning: Evidence from a Large Public Program in India*. No. 7212. IZA. <https://docs.iza.org/dp7212.pdf>.
- Alesina, Alberto, Travis Cassidy, and Ugo Troiano. 2019. “Old and Young Politicians.” *Economica* 86 (344): 689–727. <https://doi.org/10.1111/ecca.12287>.
- Amar, Priyadarshi, Apurva Bamezai, and Rithika Kumar. 2025. “Missing from the Middle: Patriarchal Constraints and the Invisibilization of Women in Everyday Governance.” <https://doi.org/10.2139/ssrn.5200785>.
- Association for Democratic Reforms. 2026. *Delhi Municipal Election Winner Profiles: 2012, 2017 and 2022*. <https://www.myneta.info/>.
- Auerbach, Adam Michael, Shikhar Singh, and Tariq Thachil. 2025. “Who Knows How to Govern? Procedural Knowledge in India’s Small-Town Councils.” *American Political Science Review* 119 (2): 708–26. <https://doi.org/10.1017/S0003055424000297>.
- Bamezai, Apurva, Siddharth George, M. R. Sharan, and Borui Sun. 2026. “The Determinants and Consequences of Political Selection in the Developing World: Evidence from India.” <https://www.mrsharan.com/papers/political-selection.pdf>.
- Ban, Radu, and Vijayendra Rao. 2007. “Tokenism or Agency? The Impact of Women’s Reservations on Village Democracies in South India.” https://vijayendraraio.org/papers/EDCC_Reservations_to_go.pdf.
- Beaman, Lori, Raghavendra Chattopadhyay, Esther Duflo, Rohini Pande, and Petia Topalova. 2009. “Powerful Women: Does Exposure Reduce Bias?” *Quarterly Journal of Economics* 124 (4): 1497–540. <https://www.povertyactionlab.org/sites/default/files/research-paper/310%20Female%20Quotas%20Nov%2009.pdf>.
- Besley, Timothy, Rohini Pande, and Vijayendra Rao. 2007. “Just Rewards? Local Politics and Public Resource Allocation in South India.” https://vijayendraraio.org/papers/justrewards_oct07.pdf.
- Carnes, Nicholas, and Noam Lupu. 2016. “What Good Is a College Degree? Education and Leader Quality Reconsidered.” *The Journal of Politics* 78 (1): 35–49. <https://doi.org/10.1086/683027>.
- Chattopadhyay, Raghavendra, and Esther Duflo. 2001. *Women as Policy Makers: Evidence from a India-Wide Randomized Policy Experiment*. No. 8615. National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w8615/w8615.pdf.
- Deininger, Klaus, Songqing Jin, Hari K. Nagarajan, and Fang Xia. 2011. *Does Female Reservation Affect Long-Term Political Outcomes? Evidence from Rural India*. No. 5708. World Bank. <https://documents1.worldbank.org/curated/en/595231468040603055/pdf/WPS5708.pdf>.
- Goyal, Tanushree. 2023. *Replication Data for: Local Political Representation as a Pathway to Power: A Natural Experiment in India*. Harvard Dataverse. <https://doi.org/10.7910/DVN/9XPV4I>.
- Green, Donald P., Manu Singh, and Gaurav Sood. 2026. “New Evidence on the Allocative Effects of Randomly Assigned Reservations for Women Leaders in Rural Indian Local Government.” https://github.com/in-rolls/quota_spending.

- Heinze, Alyssa R., Rachel E. Brulé, and Simon Chauchard. 2025. “Who Actually Governs? Gender Inequality and Political Representation in Rural India.” *Journal of Politics* 87 (2): 818–22. <https://doi.org/10.1086/730746>.
- Jain, Chandan, Shagun Kashyap, Rahul Lahoti, and Soham Sahoo. 2022. *Do Educated Leaders Affect Economic Development? Evidence from India*. No. 15278. IZA. <https://docs.iza.org/dp15278.pdf>.
- Joshi, Gaurav, Amit Joshi, and Abhishek Kumar Pant. 2025. “Does Education Make Local Elected Officials Perform Better? A Case of Gram Panchayats in India.” *Indian Journal of Public Administration* 71 (1): 53–61. <https://doi.org/10.1177/00195561241284449>.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2020a. “Can Gender Quotas Improve Public Service Provision? Evidence from Indian Local Government.” https://www.rochester.edu/college/faculty/alexander_lee/wp-content/uploads/2020/12/gender-performance7.pdf.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2020b. *Replication Data for: Do Gender Quotas Hurt Less Privileged Groups? Evidence from India*. Harvard Dataverse. <https://doi.org/10.7910/DVN/0QOVCH>.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2023. *Can Gender Quotas Improve Public Service Provision? Evidence from Indian Local Government*. <https://doi.org/10.7910/DVN/IO9SLQ>.
- Karekurve-Ramachandra, Varun, and Gaurav Sood. 2026. “The Limits of Electoral Gender Quotas in Rural Local Bodies.” https://gsood.com/research/papers/quota_raj.pdf.
- Mori, Yuko, R. Manjula, D. Rajasekhar, Jun Goto, and Takashi Kurosaki. 2022. “Who Becomes a Politician in a Gender Quota System? Evidence from India.” https://www.isid.ac.in/~epu/acegd2022/papers/Yuko_Mori.pdf.
- Mori, Yuko, R. Manjula, D. Rajasekhar, Jun Goto, and Takashi Kurosaki. 2025. “Who Becomes a Politician in a Gender Quota System? Evidence from India.” *World Development* 188: 106911. <https://doi.org/10.1016/j.worlddev.2024.106911>.
- Prakash, Nishith, Marc Rockmore, and Yogesh Uppal. 2019. “Do Criminally Accused Politicians Affect Economic Outcomes? Evidence from India.” *Journal of Development Economics* 141: 102370. <https://doi.org/10.1016/j.jdeveco.2019.102370>.
- Raabe, Katharina, Madhushree Sekher, and Regina Birner. 2009. *The Effects of Political Reservations for Women on Local Governance and Rural Service Provision: Survey Evidence from Karnataka*. No. 00878. International Food Policy Research Institute. <https://cgspace.cgiar.org/server/api/core/bitstreams/c85053d0-fd33-400a-b7bb-05a04874175f/content>.
- Sood, Gaurav. 2022. *Local-Elections: Indian Local Election Data*. GitHub repository in-rolls/local_elections. https://github.com/in-rolls/local_elections/tree/5f2261a57435c6029a15697b96432012ee20b5de/data/delhi.
- State Election Commission, NCT of Delhi. 2012. *Notification Regarding Reservation of Seats, 2012*. https://sec.delhi.gov.in/sites/default/files/SEC/circulars-orders/notification-regarding-reservation-seats-2012_1.pdf.
- State Election Commission, NCT of Delhi. 2022. *Reservation Order for the Municipal Corporation of Delhi*. https://sec.delhi.gov.in/sites/default/files/SEC/generic_multiple_files/reservationorder_0.pdf.
- Vaidyanathan, R. 2023. *MCD Election Results 2022*. OpenCity. <https://data.opencity.in/dataset/delhi->

mcd-elections-2022.

A Measurement and reproducibility

Table A.1: Education coverage among eligible winner records. N includes unclassified qualifications.

State	Year	Office	Seat	Classified (%)	N
UP	2010	Village head	Open	99.6	34,654
UP	2010	Village head	Women	99.4	17,166
UP	2015	Village head	Open	100.0	39,093
UP	2015	Village head	Women	100.0	19,926
UP	2021	Village head	Open	95.3	32,997
UP	2021	Village head	Women	94.1	16,774
Rajasthan	2020	Village head	Open	81.8	5,695
Rajasthan	2020	Village head	Women	56.9	5,433
Kerala	2010	Block member	Open	91.0	983
Kerala	2010	Block member	Women	93.0	1,095
Kerala	2010	Ward member	Open	89.7	8,047
Kerala	2010	Ward member	Women	90.8	8,594
Kerala	2010	District member	Open	87.7	162
Kerala	2010	District member	Women	94.7	169
Kerala	2015	Block member	Open	87.4	1,021
Kerala	2015	Block member	Women	88.7	1,041
Kerala	2015	Ward member	Open	87.6	7,725
Kerala	2015	Ward member	Women	89.7	8,196
Kerala	2015	District member	Open	86.0	164
Kerala	2015	District member	Women	92.2	167
Kerala	2020	Block member	Open	88.7	980
Kerala	2020	Block member	Women	89.3	1,098
Kerala	2020	Ward member	Open	83.8	7,726
Kerala	2020	Ward member	Women	84.9	8,226
Kerala	2020	District member	Open	86.1	166
Kerala	2020	District member	Women	84.2	165

Education is missing for many winners in Rajasthan, where qualifications are often recorded as “Other”, and is classified at different rates across Kerala’s seat types. Ages outside 21 to 100 are set to missing. Some Uttar Pradesh inputs come from integration work not yet in a published release of the election data; the repository keeps their byte hashes and the adapter diff.

The missing-education bounds keep the all-winner regression design and let each unclassified winner be either a graduate or not. After reservation is residualized on the controls, the coefficient is a weighted sum of outcomes, so assigning each missing outcome by the sign of its weight gives the exact minimum and maximum. The bounds carry no sampling uncertainty. Uttar Pradesh’s bounds are negative in each election; Rajasthan’s and Kerala’s include both signs.

Table A.2: Missing-education bounds on the adjusted all-winner graduate-share difference (percentage points). These are not confidence intervals.

State	Year	Office	Lower	Upper	Missing	N
UP	2010	Village head	-13.5	-12.4	254	51,820
UP	2015	Village head	-9.1	-9.1	0	59,019
UP	2021	Village head	-12.0	-1.3	2,543	49,771
Rajasthan	2020	Village head	-29.7	31.5	3,380	11,128
Kerala	2010	Block member	-12.3	3.4	165	2,078
Kerala	2010	Ward member	-10.6	8.9	1,617	16,641
Kerala	2010	District member	-12.7	4.8	29	331
Kerala	2015	Block member	-13.8	10.1	247	2,062
Kerala	2015	Ward member	-12.2	10.5	1,800	15,921
Kerala	2015	District member	-18.0	3.7	36	331
Kerala	2020	Block member	-14.5	7.5	228	2,078
Kerala	2020	Ward member	-14.4	16.9	2,495	15,952
Kerala	2020	District member	-15.1	15.2	49	331

B Complete rural estimates

B.1 Bihar: education

Table B.1: Bihar rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2016	Village head	-15.0 [-16.5, -13.4]	1.5 [0.9, 2.1]	7,873	523	Block
2016	Ward member	-4.6 [-4.9, -4.2]	11.5 [10.8, 12.3]	97,200	517	Block
2016	Sarpanch	-12.1 [-13.4, -10.8]	4.7 [3.7, 5.6]	7,841	521	Block
2016	Panch	-2.8 [-3.1, -2.6]	14.8 [14.0, 15.7]	85,355	516	Block
2016	Block member	-10.8 [-12.0, -9.5]	3.2 [2.5, 4.0]	10,743	519	Block
2016	District member	-12.7 [-18.4, -6.9]	1.3 [-0.4, 2.9]	888	38	District

B.2 Uttar Pradesh: education

Table B.2: Uttar Pradesh rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Village head	-13.1 [-14.5, -11.7]	6.3 [5.1, 7.4]	51,566	72	District
2015	Village head	-9.1 [-9.7, -8.5]	9.3 [8.7, 10.0]	59,019	823	Block
2021	Village head	-7.4 [-8.1, -6.7]	8.2 [7.5, 9.0]	47,228	728	Block

B.3 Rajasthan: education

Table B.3: Rajasthan rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2020	Village head	-6.9 [-9.0, -4.9]	0.1 [-0.1, 0.2]	7,748	352	Block

B.4 Kerala: education

Table B.4: Kerala rural officials: two parts of the education distribution. Differences in percentage points with pointwise 95% confidence intervals. N: classified winners; G: geographic clusters. Caste controls included. Illiteracy is unavailable.

Year	Office	Graduate or above	Illiterate	N	G	Controls
2010	Ward member	-0.5 [-2.4, 1.3]	—	15,024	14	District
2010	Block member	-4.1 [-7.9, -0.3]	—	1,913	150	Block
2010	District member	-4.3 [-14.1, 5.6]	—	302	14	District
2015	Ward member	-0.1 [-1.6, 1.3]	—	14,121	14	District
2015	Block member	-1.2 [-5.0, 2.6]	—	1,815	148	Block
2015	District member	-7.8 [-21.1, 5.4]	—	295	14	District
2020	Ward member	2.0 [-0.8, 4.7]	—	13,457	14	District
2020	Block member	-3.5 [-7.5, 0.4]	—	1,850	149	Block
2020	District member	-1.8 [-15.6, 12.1]	—	282	14	District

B.5 Age

Table B.5: Age of rural officials: differences in years with pointwise 95% confidence intervals. N: winners with age recorded; G: geographic clusters. Controls match the education models.

State	Year	Office	Difference [95% CI]	N	G
Bihar	2016	Village head	-3.6 [-4.1, -3.2]	7,934	524
Bihar	2016	Ward member	-2.0 [-2.2, -1.8]	97,211	517
Bihar	2016	Sarpanch	-5.6 [-6.2, -5.1]	7,871	521
Bihar	2016	Panch	-2.9 [-3.1, -2.7]	85,376	516
Bihar	2016	Block member	-2.7 [-3.1, -2.3]	10,775	519
Bihar	2016	District member	-4.2 [-5.6, -2.7]	891	38
Bihar	2021	Village head	-3.2 [-3.6, -2.7]	8,050	533
Bihar	2021	Ward member	-0.5 [-0.6, -0.3]	109,501	533
Bihar	2021	Sarpanch	-6.2 [-6.7, -5.7]	8,043	533
Bihar	2021	Panch	-3.2 [-3.4, -3.0]	106,636	533
Bihar	2021	Block member	-2.3 [-2.7, -1.9]	10,972	533
Bihar	2021	District member	-3.9 [-5.0, -2.8]	1,144	38
Uttar Pradesh	2010	Village head	-1.4 [-1.6, -1.2]	51,662	72
Uttar Pradesh	2021	Village head	-1.3 [-1.5, -1.1]	49,757	728

State	Year	Office	Difference [95% CI]	N	G
Rajasthan	2020	Village head	-1.7 [-2.3, -1.1]	11,113	352
Kerala	2010	Ward member	-4.8 [-5.9, -3.8]	14,793	14
Kerala	2010	Block member	-4.9 [-5.8, -4.0]	1,980	151
Kerala	2010	District member	-4.2 [-7.3, -1.2]	330	14
Kerala	2015	Ward member	-4.2 [-5.2, -3.2]	15,724	14
Kerala	2015	Block member	-3.9 [-4.9, -3.0]	2,029	151
Kerala	2015	District member	-3.0 [-5.8, -0.3]	327	14
Kerala	2020	Ward member	-3.5 [-4.5, -2.4]	15,860	14
Kerala	2020	Block member	-3.1 [-4.0, -2.2]	2,070	152
Kerala	2020	District member	-3.6 [-6.0, -1.2]	331	14

C Occupation, assets and criminal records

Table C.1: Occupation, assets and criminal records: reserved minus open seats, with the controls and clustering of the education models. No occupation and criminal record in percentage points; log declared assets in log points. No occupation counts nil, unemployed, homemaker and student entries. Mumbai, which records no occupation, reports whether the councillor declared a tax ID (PAN).

Outcome	State	Year	Office	Difference [95% CI]	N
No occupation	Delhi	2012	Councillor	28.0 [18.4, 37.5]	231
No occupation	Delhi	2017	Councillor	25.6 [17.1, 34.0]	264
No occupation	Delhi	2022	Councillor	21.7 [13.5, 29.9]	246
No occupation	Kerala	2010	Block member	38.8 [33.6, 43.9]	1,690
No occupation	Kerala	2010	Ward member	54.5 [50.2, 58.8]	13,992
No occupation	Kerala	2010	District member	12.4 [0.9, 23.8]	241
No occupation	Kerala	2015	Block member	35.3 [30.4, 40.3]	1,654
No occupation	Kerala	2015	Ward member	50.2 [44.9, 55.4]	12,647
No occupation	Kerala	2015	District member	15.1 [6.7, 23.6]	240
No occupation	Kerala	2020	Block member	30.7 [25.5, 35.9]	1,637
No occupation	Kerala	2020	Ward member	41.0 [35.7, 46.2]	12,267
No occupation	Kerala	2020	District member	0.2 [-10.3, 10.7]	217
No occupation	Rajasthan	2020	Village head	51.1 [48.2, 53.9]	11,128
No tax ID declared	Mumbai	2012, 2017	Councillor	5.8 [2.1, 9.4]	449
Log declared assets	Rajasthan	2020	Village head	-0.6 [-0.8, -0.5]	11,128
Log declared assets	Uttar Pradesh	2021	Village head	-0.1 [-0.1, -0.0]	49,771
Criminal record	Uttar Pradesh	2021	Village head	-0.0 [-0.1, 0.0]	42,720

D Delhi estimates and coverage

Table D.1: Delhi councillors: adjusted differences in percentage points, except age in years. Assembly-constituency and caste-reservation controls; pointwise intervals clustered by assembly constituency. G denotes clusters.

Year	Outcome	Difference [95% CI]	N	G
2012	Graduate or above	-0.6 [-14.4, 13.2]	226	65
2012	Illiterate	2.5 [-1.3, 6.3]	226	65
2012	Age (years)	-6.0 [-9.0, -3.0]	235	65
2012	Any pending case	-23.5 [-35.2, -11.8]	237	65
2012	no_earnings	28.0 [18.4, 37.5]	231	64
2012	no_earnings_social_missing	30.9 [20.7, 41.1]	213	62
2017	Graduate or above	1.1 [-10.4, 12.6]	262	68
2017	Illiterate	0.6 [-1.7, 2.8]	262	68
2017	Age (years)	-3.6 [-5.6, -1.5]	262	68
2017	Any pending case	-10.6 [-18.4, -2.8]	267	68
2017	no_earnings	25.6 [17.1, 34.0]	264	68
2017	no_earnings_social_missing	27.0 [18.1, 35.8]	245	68
2022	Graduate or above	-3.4 [-15.9, 9.1]	244	68
2022	Illiterate	2.4 [-1.0, 5.7]	244	68
2022	Age (years)	-2.3 [-4.8, 0.3]	248	68
2022	Any pending case	-21.0 [-29.9, -12.0]	248	68
2022	no_earnings	21.7 [13.5, 29.9]	246	68
2022	no_earnings_social_missing	24.7 [15.5, 33.9]	226	68

Table D.2: Delhi outcome coverage among recorded winners. Percentages use all winners of each seat type as the denominator.

Year	Seat	N	Education (%)	Age (%)	Cases (%)
2012	Open	134	85.8	86.6	87.3
2012	Women	138	80.4	86.2	87.0
2017	Open	134	97.0	98.5	99.3
2017	Women	138	95.7	94.2	97.1
2022	Open	125	97.6	100.0	100.0
2022	Women	125	97.6	98.4	98.4

Table D.3: Delhi missing-outcome bounds on the adjusted all-winner coefficient, in percentage points. They carry no sampling uncertainty.

Year	Outcome	Lower	Upper	Missing	N
2012	Graduate or above	-16.4	18.9	46	272
2012	Any pending case	-33.9	-6.1	35	272
2017	Graduate or above	-2.5	6.3	10	272
2017	Any pending case	-11.5	-7.5	5	272

Year	Outcome	Lower	Upper	Missing	N
2022	Graduate or above	-5.0	-0.1	6	250
2022	Any pending case	-21.3	-19.9	2	250

E Published comparisons

Table E.1: Published comparisons of leader qualifications by women’s reservation

Study	Setting	Design	Outcome	Reserved	Open	Difference (SE)	N
<i>Elected village leaders, women-reserved vs. open seats</i>							
Chattopadhyay and Duflo (2001) <i>Table 3, panels A and B</i>	West Bengal (Birbhum); 1998 term; GP pradhan	Randomized rotation	Years of schooling (years)	7.13	9.92	−2.79 (0.54)	161
			Literate (%)	80.0	98.0	−19.0 (4.0)	161
			Elected to GP council before 1998 (%)	11.0	43.0	−32.0 (7.0)	161
Beaman et al. (2009) <i>Table II, column 6, printed p. 1507</i>	West Bengal (Birbhum); 2003 term; GP pradhan	Randomized rotation; adjusted	Years of schooling (years)			−2.104 (0.548)	165
			First time as pradhan in 2003 term (%)			26.5 (6.8)	165
			Other political experience (%)			−12.1 (7.9)	165
Ban and Rao (2007) <i>Table 8 (means), Table 9 column 2 (difference)</i>	Andhra Pradesh, Karnataka, Kerala, Tamil Nadu; 2002 survey; GP president	Rotation; caste-reserved GPs excluded; adjusted	Years of schooling (years)	6.074	8.456	−2.620 (0.766)	106
			Political knowledge score	3.185	4.608	−1.413 (0.292)	106
			Prior political positions (count)			−0.461 (0.188)	106
Deininger et al. (2011) <i>Table 3, printed p. 22</i>	National (REDS, 233 villages); Up to three elections; REDS 2007 survey; GP pradhan	Descriptive	Illiterate (%)	24.7	5.7	19.0 (3.4) [†]	639
			At least secondary education (%)	28.6	62.7	−34.0 (4.1) [†]	639
			Held political office before (%)	11.7	24.2	−12.6 (3.1) [†]	639
Afridi, Iversen and Sharan (2013) <i>Table 2, panel I, printed p. 38</i>	Andhra Pradesh (8 districts); In office 2006–10; GP sarpanch	Descriptive	Illiterate (%)	9.0	0.0	9.0 (2.2)	297
			Higher secondary or more (%)	15.0	54.0	−40.0 (5.2)	297
			Own prior political experience (%)	11.0	22.0	−11.0 (4.4)	297
Bamezai et al. (2026) <i>Table 7, panel B</i>	Bihar; 2016 and 2021 elections; Mukhiya	Fuzzy RD at population threshold; adjusted	Schooling, SD of all citizens			−0.921 (0.073)	3,294
<i>Elected leaders, reserved women vs. men in other seats</i>							
Raabe, Sekher and Birner (2009) <i>Table 4</i>	Karnataka (12 districts); 2006 survey; GP president	Descriptive	Illiterate (%)	9.0	4.0	5.0 (6.6) [†]	71
			Degree or above (%)	0.0	19.0	−19.0 (5.7) [†]	71
<i>Municipal councillors, women-reserved vs. open seats</i>							

Table E.1 (continued)

Study	Setting	Design	Outcome	Reserved	Open	Difference (SE)	N
Karekurve-Ramachandra and Lee (2020) <i>Table 1, printed p. 14</i>	Mumbai; 2011–2018 (2012 and 2017 terms); Municipal councillor	Descriptive	Attended university (%)	43.7	39.0	4.7 (2.9) [†]	1,118
			Criminal record (%)	3.7	30.3	−26.6 (2.1) [†]	1,118
<i>Candidates, women-reserved vs. open seats</i>							
Mori et al. (2022) <i>Table 3, panel B, column 1</i>	Karnataka (Kolar, Tumkur); December 2020 election; GP member (single-seat)	Random rotation; adjusted	Education position among all villagers			−0.41 (0.33)	97
Bamezai et al. (2026) <i>Table 7, panel A</i>	Bihar; 2016 and 2021 elections; Mukhiya	Fuzzy RD at population threshold; adjusted	Schooling, SD of all citizens			−0.765 (0.024)	44,583
<i>Elected leaders, other benchmarks</i>							
Mori et al. (2022) <i>Table 3, panel B, column 3</i>	Karnataka (Kolar, Tumkur); December 2020 election; GP member (single-seat)	Random rotation; adjusted	Education position among same-gender villagers <i>Reserved winner's position among women vs. open winner's position among men</i>			0.74 (0.44)	92
Bamezai et al. (2026) <i>Table 7, panel B</i>	Bihar; 2016 and 2021 elections; Mukhiya	Fuzzy RD at population threshold; adjusted	Schooling, SD of same-gender citizens <i>Reserved winner's position among women vs. open winner's position among men</i>			−0.619 (0.074)	3,249
			Husband's schooling, SD of all citizens <i>Reserved winner's husband vs. open-seat winner</i>			−0.364 (0.089)	3,058

computed from shares and group sizes as for independent samples, or from the reported t-statistic; these ignore clustering and repeated units. N is the number of reserved plus open units where reported separately, otherwise the regression sample. SD rows are standardized against the GP's citizens; selection-score rows measure position in the villager education distribution. Chattopadhyay and Duflo (2001) and Beaman et al. (2009) study the same Birbhum GPs in successive terms; the published version of Chattopadhyay and Duflo (2004) omits the pradhan-characteristics table. Karekurve-Ramachandra and Lee (2020) use the same Mumbai councillor data as this paper. Mori et al.'s published version (2025) reports the same-gender benchmark. Row-level source locations and transcription notes are in the repository's lit/studies files.